12TH BIOLOGY · CHAPTER 25 · FULL CHAPTER NOTES

# Ecology

Topic-wise short and long questions with answers from the complete chapter

## 25.1 LEVELS OF ECOLOGICAL ORGANIZATION

### Q.1 [Short] Define ecology.

Ecology is the study of the relationships of organisms with other organisms and with the environment. Organisms fulfil their requirements for life by these interactions.

### Q.2 [Long] Describe the levels of ecological organization.

| Level | Definition | Example from Pakistan |
|---|---|---|
| 1. Species | A group of organisms that share similar physical and genetic characteristics and can interbreed to produce fertile offspring; the basic unit of classification | Indus River Dolphin |
| 2. Population | A group of individuals of the same species living in the same geographic area at the same time | All the Markhors of Chitral Gol National Park |
| 3. Community | All the different populations of various species living and interacting in the same area | Mangrove forest of Karachi: mangrove trees, mudskipper fish, crabs, shorebirds |
| 4. Ecosystem | The biotic (living) and abiotic (non-living) components of the environment together; energy flows unidirectionally and nutrients cycle | Thar Desert: snakes, blackbucks, shrubs plus heat, sand dunes, rare rainfall |
| 5. Biome | A large, distinct geographical region characterized by its specific climate, soil, altitude, and types of plants and animals adapted to it | Coniferous forest biome (Murree, Swat); desert biome (Thal, Thar); freshwater biome (Indus River system) |
| 6. Biosphere | The highest and most inclusive level; the global sum of all biomes; every part of the Earth where life exists | From the trenches of the Arabian Sea to the peaks of the Himalayas |

### Q.3 [Short] Differentiate between a species and a population.

A species refers to a “type” of organism globally, while a population refers to a specific group in a specific location. Markhors across all of Pakistan are the species; those living in Chitral Gol National Park form one population.

### Q.4 [Short] Differentiate between a community and an ecosystem.

A community consists only of the living populations of an area (e.g., snakes, blackbucks and shrubs of the Thar Desert). An ecosystem includes the community plus the abiotic components such as sunlight, soil, water and air (e.g., the baking heat, sand dunes and rare rainfall of the Thar).

### Q.5 [Short] Why is the biosphere called the “global ecosystem”?

The biosphere is the global sum of all biomes. It is a thin “envelope” around the Earth where the lithosphere (land), hydrosphere (water) and atmosphere (air) interact to support life.

## 25.2 BIOGEOCHEMICAL CYCLES

### Q.6 [Short] What are biogeochemical cycles?

**Biogeochemical cycles** are the cyclic movements of chemical elements, such as carbon, nitrogen, oxygen and phosphorus, between the living (biotic) and non-living (abiotic) components of the ecosystem.

### Q.7 [Short] What is a primary reservoir? Classify biogeochemical cycles on its basis.

A **primary reservoir** (“sink” or “pool”) is a massive component of the biosphere where the element of a biogeochemical cycle is stored for long periods of time. On this basis the cycles are classified as:

- **Gaseous cycles:** the reservoir is the air or the oceans (via evaporation). They include the nitrogen, oxygen, carbon and water cycles.
- **Sedimentary cycles:** the reservoir is the Earth’s crust. They include the phosphorus, sulfur and calcium cycles.

### Q.8 [Long] Describe the water cycle in detail.

The **water cycle** (hydrological cycle) is the continuous movement of water between the abiotic components (air, soil and sea) and the biotic components.

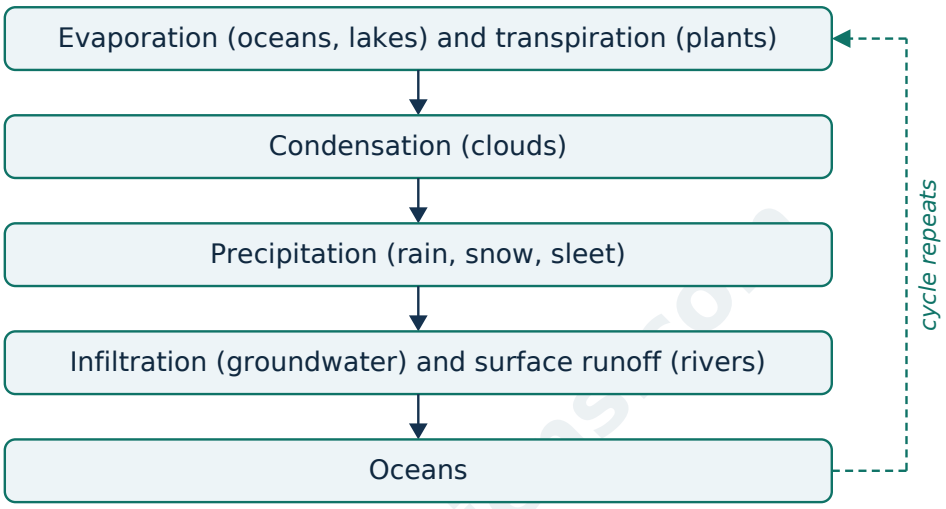
#### Abiotic loop

1. **Evaporation:** The sun powers the entire cycle. Solar radiation heats the surface water of oceans and lakes, and water molecules evaporate as vapours. About 86% of global evaporation occurs from the oceans (for Pakistan, the Arabian Sea).
2. **Condensation:** Rising vapours lose heat. When the air is cold enough, they turn back into liquid droplets or ice crystals, which cling to dust particles to form clouds.
3. **Precipitation:** When the droplets become too heavy, they fall due to gravity as rain, snow or sleet (rain mingled with snow or hail).
4. **Infiltration and runoff:** In *infiltration*, water seeps into the soil (percolation) and moves down to the water table, where it is stored as groundwater. *Surface runoff* is the unconfined flow of water over the land when it can no longer infiltrate; it flows into streams, rivers and finally the oceans.

studysapiens.com

## Biotic loop

1. **Entry of water:** Terrestrial plants absorb water from the soil through root hairs and use it for photosynthesis and for support (turgor pressure). Animals take in water by drinking and by eating food.
2. **Exit of water:** *Transpiration* through the stomata of leaves is the major exit point. *Respiration* in plants and animals produces "metabolic water", exhaled as vapours. In *excretion and egestion*, animals release water in urine and faeces. In *perspiration*, water leaves through skin pores and evaporates. *Decomposition* of dead bodies also returns water to the soil or atmosphere.




*The water cycle*

### Q.9 [Short] Differentiate between infiltration and surface runoff.

| Infiltration | Surface runoff |
|---|---|
| Water seeps into the soil (percolation) and moves down to the water table. | Unconfined flow of water over the land surface when it can no longer infiltrate. |
| Stores water underground as groundwater; recharges aquifers. | Flows into streams, rivers and finally the oceans; feeds the Indus River. |

### Q.10 [Short] How does water leave the biotic components?

By transpiration through the stomata of leaves (the major exit), respiration (metabolic water exhaled as vapours), excretion and egestion (urine and faeces), perspiration through skin pores, and decomposition of dead bodies.

### Q.11 [Short] What is metabolic water?

Respiration in plants and animals produces water as a by-product. This "metabolic water" is exhaled as vapours.

## Q.12 [Long] Describe the nitrogen cycle in detail.

The **nitrogen cycle** is the process by which nitrogen circulates between the biotic and abiotic components of the ecosystem. Atmospheric nitrogen ($N_2$) is transformed into compounds that living organisms can use, and is later returned to the atmosphere. It has five steps:

### 1. Nitrogen fixation (entry into the soil)

The atmosphere is 78% nitrogen gas, whose atoms are joined by a strong triple bond that plants cannot break. Nitrogen fixation is the conversion of $N_2$ into inorganic compounds like ammonia ($NH_3$). It occurs in three ways:

- **Biological fixation** (more than 90%): by soil microorganisms. *Free-living* (non-symbiotic): cyanobacteria (blue-green algae), *Azotobacter*, *Clostridium*. *Mutualistic* (symbiotic): *Rhizobium* in the root nodules of legumes such as chickpea (chana) and mung bean (moong). They convert $N_2$ to ammonia.
- **Atmospheric fixation** (5–10%): lightning during rains and storms gives the energy to combine nitrogen with oxygen; $N_2$ is converted into nitrogen oxides and then into nitrates.
- **Industrial fixation:** the synthesis of nitrogen-containing fertilizers; atmospheric nitrogen and hydrogen are combined to produce ammonia.

### 2. Nitrification (conversion to plant food)

Ammonia is toxic or difficult for plants to use in large amounts. Soil bacteria convert it: *Nitrosomonas* convert ammonia to nitrite ($NO_2^-$), and *Nitrobacter* convert nitrite to nitrate ($NO_3^-$). Nitrates are easily absorbed by plants.

### 3. Assimilation (entry into the biotic world)

Plants and microbes absorb nitrates from the soil and use them to make proteins, DNA and chlorophyll. Animals assimilate nitrogen by eating plants and other animals.

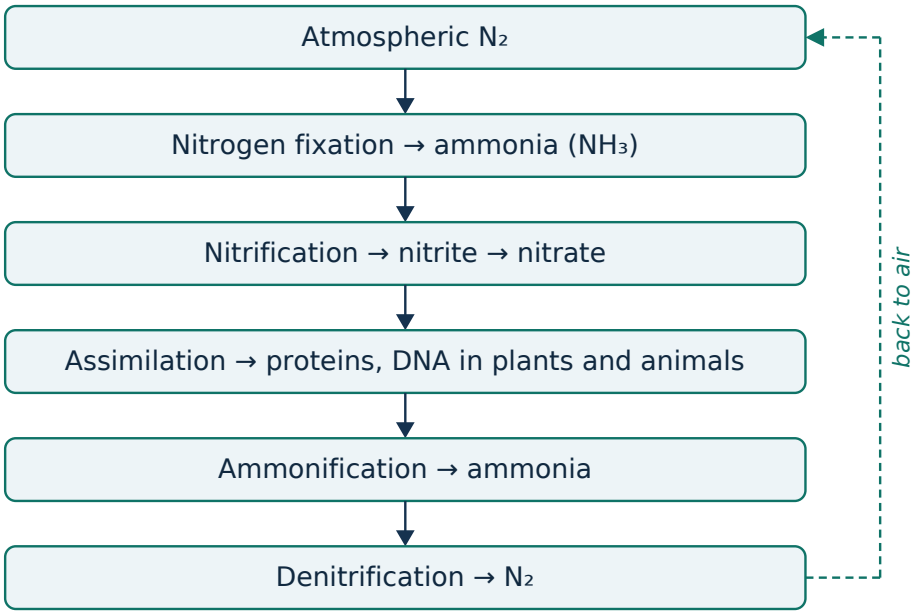
### 4. Ammonification (the recycling step)

When plants and animals die, or when animals release nitrogenous waste, decomposers (fungi and bacteria) break down the complex proteins and convert the organic nitrogen back into ammonia.

### 5. Denitrification (exit to the atmosphere)

In waterlogged soils, denitrifying bacteria (e.g., *Pseudomonas* and *Thiobacillus*) convert nitrates back into nitrogen gas ($N_2$), which is released into the atmosphere. This keeps the cycle balanced.

studysapiens.com




*The nitrogen cycle*

### Q.13 [Short] Why can plants not use atmospheric nitrogen directly?

The atmosphere is 78% nitrogen gas ($N_2$), but its atoms are joined by a strong triple bond. Plants cannot break this bond; nitrogen must first be fixed into compounds like ammonia and nitrates.

### Q.14 [Short] Name the bacteria of the nitrogen cycle with their roles.

| Bacteria | Step | Role |
|---|---|---|
| Cyanobacteria, *Azotobacter*, *Clostridium* (free-living) | Nitrogen fixation | Convert $N_2$ to ammonia |
| *Rhizobium* (symbiotic, in root nodules of legumes) | Nitrogen fixation | Convert $N_2$ to ammonia |
| *Nitrosomonas* | Nitrification | Convert ammonia to nitrite |
| *Nitrobacter* | Nitrification | Convert nitrite to nitrate |
| Decomposer bacteria (with fungi) | Ammonification | Convert organic nitrogen to ammonia |
| *Pseudomonas*, *Thiobacillus* | Denitrification | Convert nitrates to $N_2$ |

### Q.15 [Short] What are the three ways of nitrogen fixation?

(i) **Biological fixation** by soil microorganisms (more than 90%); (ii) **atmospheric fixation** by lightning (5–10%); (iii) **industrial fixation**, the synthesis of nitrogen-containing fertilizers from atmospheric nitrogen and hydrogen.

## 25.3 THE FLOW OF ENERGY

### Q.16 [Long] What are trophic levels? Describe them.

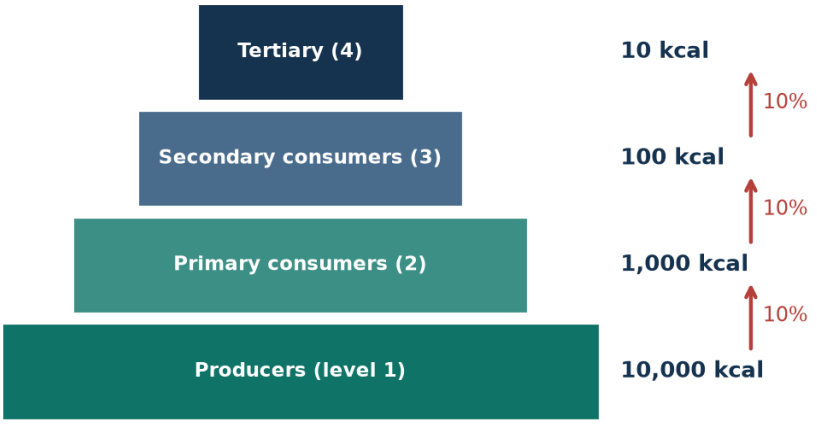
In an ecosystem the organisms are arranged in different feeding groups, known as **trophic levels**.

| Trophic level | Organisms | Examples |
|---|---|---|
| 1 | Primary producers | Green plants, algae, photosynthetic bacteria |
| 2 | Primary consumers (herbivores); feed directly on producers | Insects, rabbit |
| 3 | Secondary consumers (carnivores) and parasites of animals; feed on herbivores | Frog, snake |
| 4 | Tertiary consumers (animals that eat carnivores) | Wolf, hawk |
| Separate level | Decomposers and detritivores; feed on all trophic levels (detritus and faecal matter) | Crabs, vultures, jackals |

### Q.17 [Long] Discuss the loss of energy between trophic levels.

Unlike the cycling of matter, the flow of energy is **unidirectional**. There are generally three or four trophic levels in an ecosystem.

- About **1 to 2%** of the solar energy that falls on the producers is converted to the bond-energy of organic matter.
- Only about **10%** of the energy of one trophic level is transferred to the next; energy is lost at each level.
- So very little usable energy remains after it has passed through the bodies of organisms at different trophic levels.
- **Example:** If producers fix 10,000 kcal, the primary consumers get only 1,000 kcal, the secondary consumers about 100 kcal, and the tertiary consumers about 10 kcal.




*Loss of energy between trophic levels (textbook example)*

### Q.18 [Short] Why are short food chains more energy efficient than long food chains?

Short food chains (fewer trophic levels) are more energy efficient than long ones, because energy is lost as heat during transfer at each trophic level.

### Q.19 [Short] What are detritivores?

Decomposers and detritivores (crabs, vultures and jackals) make a separate trophic level. They feed on all trophic levels, consuming detritus (dead, decomposing plant and animal material) and faecal matter.

## 25.4 CHARACTERISTICS OF POPULATION

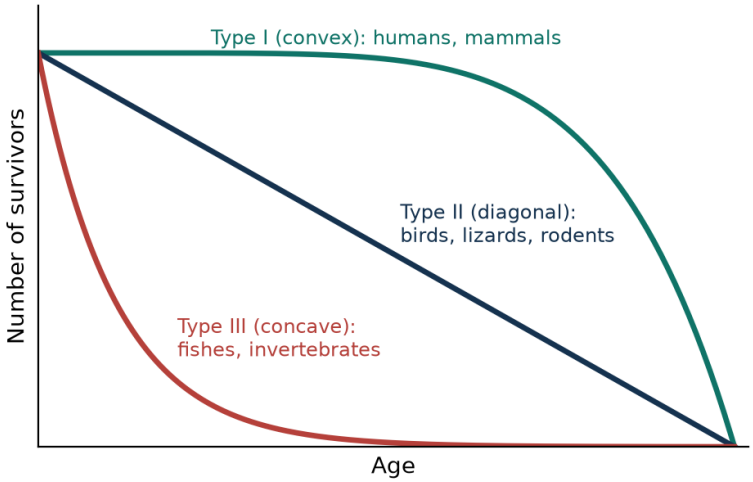
### Q.20 [Short] What is demography?

Demography is the statistical study of the characteristics of populations. Population size is its most important feature, because it is directly related to the ability of a population to survive.

### Q.21 [Long] What is population survivorship? Describe the three types of survivorship curves.

**Population survivorship** is the percentage of members of a group that are alive at a given age. It is shown by survivorship curves (Y-axis: number of survivors; X-axis: age). There are three kinds:

| Type | Pattern of death | Examples |
|---|---|---|
| Type I (convex) | Individuals survive to an old age, then die rapidly; environmental factors have very little impact on mortality | Humans and mammals |
| Type II (diagonal) | Constant probability of death throughout life; the environment is no harsher on the young than on the old | Birds, lizards, rodents |
| Type III (concave) | Very high juvenile mortality; those reaching adulthood have a much lower mortality rate | Fishes and many invertebrates |



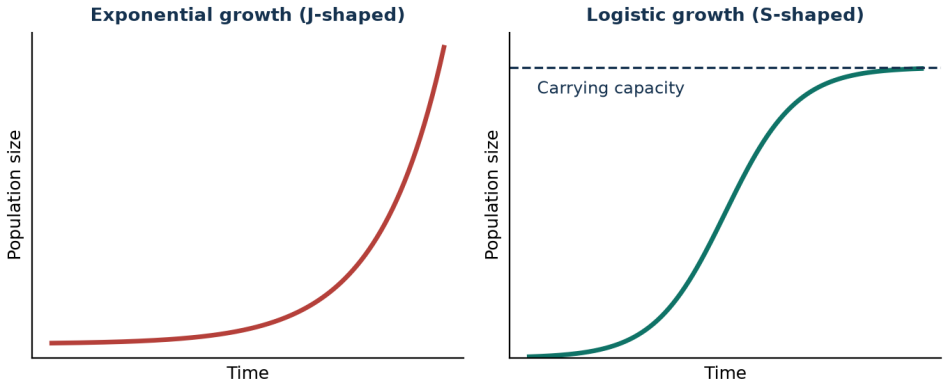

*Survivorship curves*

*The textbook numbers the three types as 1, 1, 2 (p.236); they are Type I, II and III.*

## Q.22 [Long] Describe the models of population growth.

**Population growth** is the increase in the number of individuals over a specific time period. There are two growth models:

- **Exponential growth model:** the population grows without limits at its maximal rate. It does not add a constant number each generation; it increases by the same ratio per unit time. Not all populations show it.
- **Logistic growth model:** the growing population eventually reaches a limit due to shortage of resources (space, light, water, nutrients). It stabilizes at a certain size, the **carrying capacity** of the place. The growth curve is sigmoid (flattened S).




*Population growth models*

## Q.23 [Short] What is population density?

**Population density** is the number of individuals per unit area or volume, e.g., the number of mulberry trees per square kilometre in Muzaffarabad, or the number of *E. coli* per millilitre in a test tube. At high density, as a population approaches its carrying capacity, competition for resources becomes severe; the birth rate decreases and the risk of mortality increases. High densities can also cause high rates of pollution.

### Q.24 [Long] Describe the patterns of population dispersion (distribution).

| Spacing | Description | Examples |
|---|---|---|
| Random | Individuals do not interact strongly with one another; not common in nature | Plants with wind-dispersed seeds that germinate wherever they fall |
| Uniform | Individuals are evenly distributed | Plants that inhibit the growth of neighbours; territorial animals such as nesting penguins |
| Clumped | Individuals form groups or clusters due to uneven distribution of resources; common in nature | Plants that drop seeds straight to the ground; animals that live in groups |

### Q.25 [Short] Why is clumped spacing common in nature?

Because resources are unevenly distributed, and individual animals, plants and microorganisms tend to prefer resourceful habitats, so they collect there in groups.

### Q.26 [Short] What is carrying capacity?

**Carrying capacity** is the maximum number of individuals of a particular species that a specific environment can support sustainably over a long period. It is determined by "limiting factors": availability of food, water, shelter and nesting sites, and the presence of predators and diseases.

### Q.27 [Short] What is minimal viable population size?

The **minimal viable population size** is the smallest number of individuals required for a population to survive and persist in the ecosystem. Below this critical size, the population faces a high risk of extinction. In a very small population (like the few remaining Himalayan Brown Bears in Pakistan) inbreeding occurs, which leads to genetic weaknesses and lower immunity. Moreover, a single natural disaster (flood or forest fire) may wipe it out.

### Q.28 [Short] What is the "Day of Eight Billion"?

The global population reached 8 billion in 2022, having taken 12 years to grow from 7 to 8 billion. It will take about 15 years (until 2037) to reach 9 billion, and by 2050 it will be nearly 10 billion, mostly in developing regions.

### Q.29 [Long] Explain the rapid growth of human population and its effects on the ecosystem.

Starting in the early 1700s, changes in technology gave humans more control over their food supply, superior weapons against predators, and remedies against many diseases. Humans also became less vulnerable to climatic uncertainties. These changes allowed humans to **expand the carrying capacity** of their habitats, so the population grew rapidly.

The world population reached 8 billion in 2022; about 16% lives in developed countries and 84% in developing countries. Pakistan's population was 241.49 million in the 2023 census, growing at 2.55% annually.

The rapidly growing human population is perhaps the greatest challenge to the future of the biosphere. Its impacts are:

- Habitat destruction and changes in land use
- Depletion of natural resources (freshwater, fossil fuels etc.)
- Poverty and inflation
- Increased waste generation and elevated pollution
- Climate change
- Deforestation and loss of biodiversity
- Increased chances of new epidemics and pandemics, starvation and malnutrition

### Q.30 [Short] Give some facts about the population of Pakistan.

The 2023 census recorded 241.49 million people, with an annual growth rate of 2.55%. According to the UNO, the population has now crossed 250 million, and Pakistan is the 5th most populous country in the world.

### Q.31 [Short] What share of the Earth's biomass is made by humans, plants and bacteria?

Humans make up only about 0.01% of the biomass on Earth. Plants make about 82% and bacteria about 13%.

## 25.5 GREENHOUSE EFFECT

### Q.32 [Short] What is the greenhouse effect?

The **greenhouse effect** is the natural process in which gases present in the Earth's atmosphere trap the heat radiated from the Earth's surface. These gases are called **greenhouse gases (GHGs)**. They allow sunlight to pass through but prevent heat from escaping directly back into space, so they keep the Earth warm enough to sustain life.

### Q.33 [Short] What is the enhanced greenhouse effect?

There has been a marked increase in the concentration of GHGs, particularly $CO_2$, mainly due to the use of fossil fuels for running machinery, driving vehicles, heating homes and generating electricity. The enhanced, unnatural warming of the atmosphere caused by excessive GHGs is the major negative impact of human activities on the environment.

### Q.34 [Short] Name the major greenhouse gases with their sources.

| Greenhouse gas | Sources | Share of global warming |
|---|---|---|
| Carbon dioxide ($CO_2$) | Burning fossil fuels (coal, oil, gas); deforestation | About 80% |
| Methane ($CH_4$) | Production of coal and gas; livestock (enteric fermentation in cattle); agricultural practices | About 11% |
| Nitrous oxide ($N_2O$) | Agricultural and industrial activities; combustion of fossil fuels | About 6% |
| Fluorinated gases (chlorofluorocarbons) | Synthetic; industrial processes (old refrigerators and air conditioners) | About 3% |

### Q.35 [Long] Describe the harmful effects of greenhouse gases on the environment.

1. **Global warming:** the long-term increase in the Earth's average surface temperature due to human activities. According to the IPCC Report (2023), global surface temperature has risen by roughly 1.1°C compared with the pre-industrial era (1850–1900).
2. **Heatwaves:** cities like Jacobabad and Sibi are among the hottest places on Earth. Above 50°C the human body cannot adjust its temperature, leading to fatal heatstroke; livestock suffer heat stress. Shorter, warmer winters let pests and mosquitoes survive longer. Heat worsens drought and increases the risk of wildfires.
3. **Floods:** a warmer atmosphere holds more moisture, giving erratic and intense rainfall and "turbo-charged" monsoons. Floods damage forests, homes and agriculture, cause soil erosion, nutrient depletion and contamination of water, and spread water-borne diseases.
4. **Glacial melting:** glaciers (like Baltoro and Batura) melt rapidly, causing Glacial Lake Outburst Floods. In the long run, once the glaciers vanish, the Indus River could run dry.
5. **Sea level rise and coastal erosion:** polar ice melts and seawater expands with heat. In Thatta and Badin, seawater intrusion is turning fertile land into salty marshes.
6. **Food insecurity:** high temperature at the flowering stage of wheat and rice reduces grain yield; intense heat evaporates soil moisture and turns fertile land into desert (desertification).
7. **Ecosystem disruption:** rapid shifts in temperature and rainfall threaten biodiversity; species are forced to change habitat or migrate, and may become threatened or extinct.

studysapiens.com

### Q.36 [Short] How has the $CO_2$ concentration of the atmosphere changed?

According to the WMO, before industrialization (1900) the $CO_2$ concentration was about 280 ppm. It has risen to 428 ppm in 2026 and continues to rise. The mean global temperature has increased about 1°C since 1900.

### Q.37 [Short] Why is climate change called an injustice for Pakistan?

Pakistan contributes less than 1% of global GHG emissions, yet it is ranked among the top 10 countries most vulnerable to the devastating effects of climate change.

### Q.38 [Short] What are a GLOF and a cloudburst?

A **Glacial Lake Outburst Flood (GLOF)** is a sudden release of water from a glacial lake due to the breaking of its natural dam (e.g., ice); the water can travel over 100 km. A **cloudburst** is a sudden and extremely heavy rainfall in a short period over a small area.

## 25.6 IMPORTANT ECOSYSTEMS OF PAKISTAN

### Q.39 [Long] Describe the temperate deciduous forest of Pakistan.

| Feature | Description |
|---|---|
| Nature | Found in moist mountain regions with distinct seasons; a cold winter causes the trees to drop their leaves (deciduous) |
| Location | Northern areas: Shogran, Neelam Valley, Murree and parts of Kaghan |
| Soil | Rich, greyish-brown soil with high organic matter (humus) |
| Climate | Moderate temperatures (4°C to 30°C); annual rainfall 750 mm to 1500 mm |
| Key flora | *Taxus baccata* (Yew), *Pinus wallichiana* (Blue Pine), *Cedrus deodara* (Himalayan Cedar); shrubs like *Berberis lyceum* (Sumbal) |
| Key fauna | Rhesus monkey (*Macaca mulatta*), black bear, leopard cat, wolves and various bird species |
| Current status | **Degraded.** Many large predators have been wiped out; pressure from the lumber industry and agricultural expansion |

### Q.40 [Long] Describe the tropical thorn forest of Pakistan.

| Feature | Description |
|---|---|
| Nature | The most widespread terrestrial ecosystem of the plains of Pakistan; xerophytic (drought-resistant) plants with thorns to deter herbivores and reduce water loss |
| Location | The Indus Plains: much of Punjab (Bahawalpur, Khanewal, Mianwali, Dera Ghazi Khan), Sindh (southern regions bordering the Indus) and Balochistan (western regions) |
| Climate | Arid to semi-arid; summer temperatures can reach 50°C |

| Feature | Description |
|---|---|
| Water | Scanty rainfall (200 mm to 750 mm annually) |
| Key flora | *Prosopis cineraria* (Jhand/Kandi), *Salvadora oleoides* (Peelu/Wan), *Capparis aphylla* (Karir), *Vachellia nilotica* (Kikar) |
| Key fauna | Camels, houbara bustard, chinkara, golden jackal, Balochistan gerbil, porcupines, rattlesnakes |
| Current status | **Heavily degraded.** Most of these "natural jungles" have been cleared for human settlements and the canal-irrigated agriculture of the Punjab |

## Q.41 [Long] Describe the alpine and sub-alpine meadows of Pakistan.

| Feature | Description |
|---|---|
| Nature | High-altitude "cold deserts" or grasslands, where the climate is too harsh for large trees |
| Location | Mountain tops of Gilgit-Baltistan, Chitral, Swat, Azad Jammu & Kashmir, and the Deosai Plains |
| Temperature | Severe cold; mean annual temperature between 1°C and 10°C |
| Water | Precipitation mostly as heavy snow (up to 6 feet) |
| Key flora | Stunted junipers, birch (*Betula utilis*), willow, alpine grasses and wildflowers |
| Key fauna | Snow leopard, Himalayan brown bear, Himalayan ibex, musk deer, marmots; birds like snowcock, monal, golden eagle |
| Current status | **Fragile.** Highly sensitive to global warming; melting snow caps alter the delicate balance of the meadow |

## Q.42 [Long] Describe the mangrove forest (coastal ecosystem) of Pakistan.

| Feature | Description |
|---|---|
| Nature | "The Forests of the Sea": salt-tolerant forests at the interface of land and sea |
| Location | Sindh (Keti Bandar and Mirpur Sakro mangroves in the Indus River Delta) and the Balochistan coast (Miani Hor, Kalmat Khor, Jiwani and Gawatar Bay) |
| Water | Brackish (a mix of fresh river water and salty seawater) |
| Soil | Anaerobic (oxygen-poor) mudflats; the trees have "breathing roots" |
| Key flora | *Avicennia marina* (Timmar) makes up 95% of the forest |
| Key fauna | Crocodiles, shrimps, mudskippers, crabs, lobsters, turtles, snakes, dolphins; birds like flamingo, crane, pelican |

| Feature | Description |
|---|---|
| Current status | **Threatened** by reduced freshwater flow from the Indus and pollution from Karachi; recent restoration projects have helped |

## Q.43 [Short] Compare the current status of the four ecosystems.

| Ecosystem | Status | Main threat |
|---|---|---|
| Temperate deciduous forest | Degraded | Lumber industry, agricultural expansion |
| Tropical thorn forest | Heavily degraded | Clearing for settlements and canal-irrigated agriculture |
| Alpine and sub-alpine meadows | Fragile | Global warming (melting snow caps) |
| Mangrove forest | Threatened | Reduced freshwater flow of the Indus; pollution from Karachi |

studysapiens.com